



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to BALA MURUGAN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

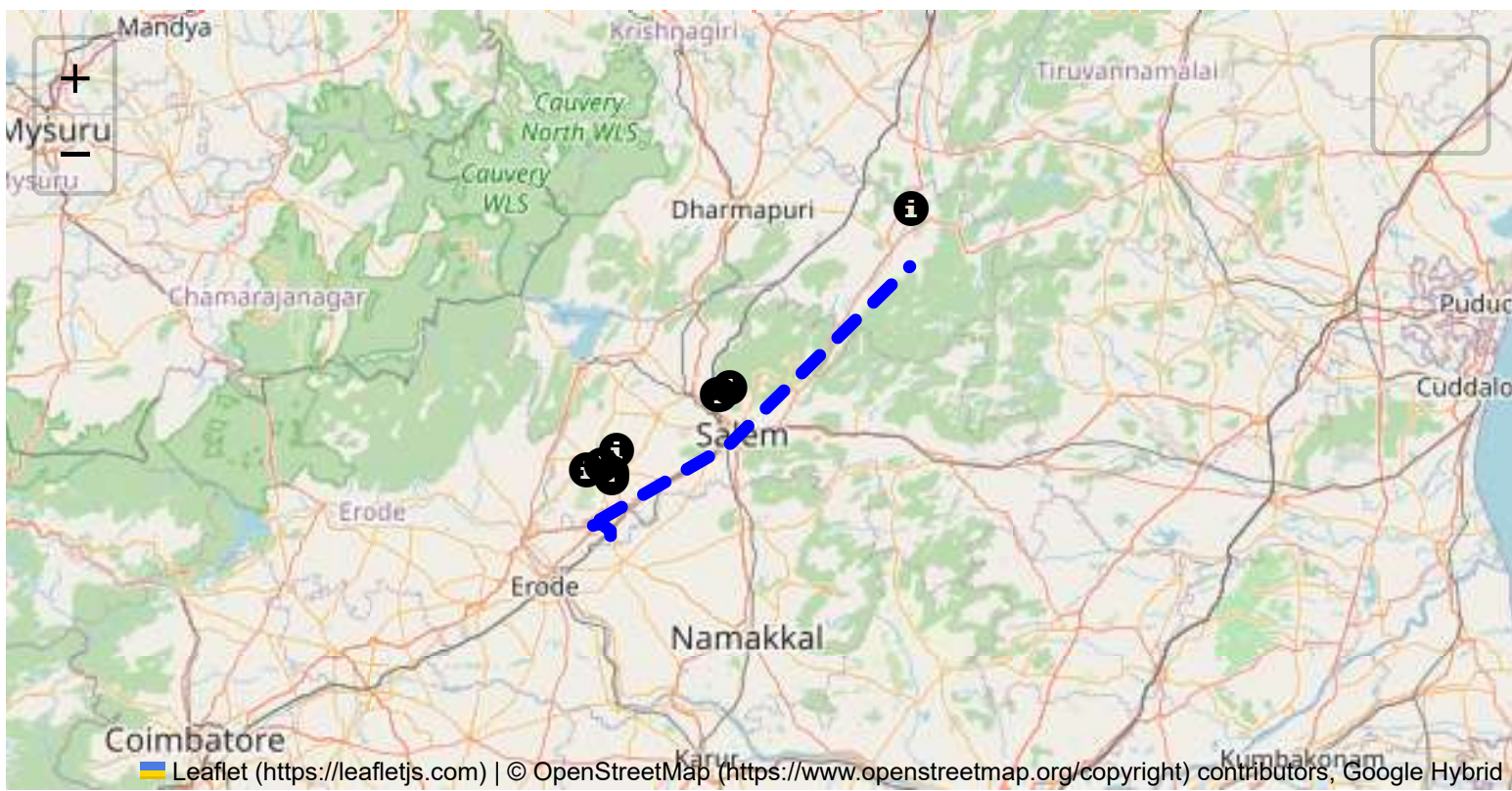
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

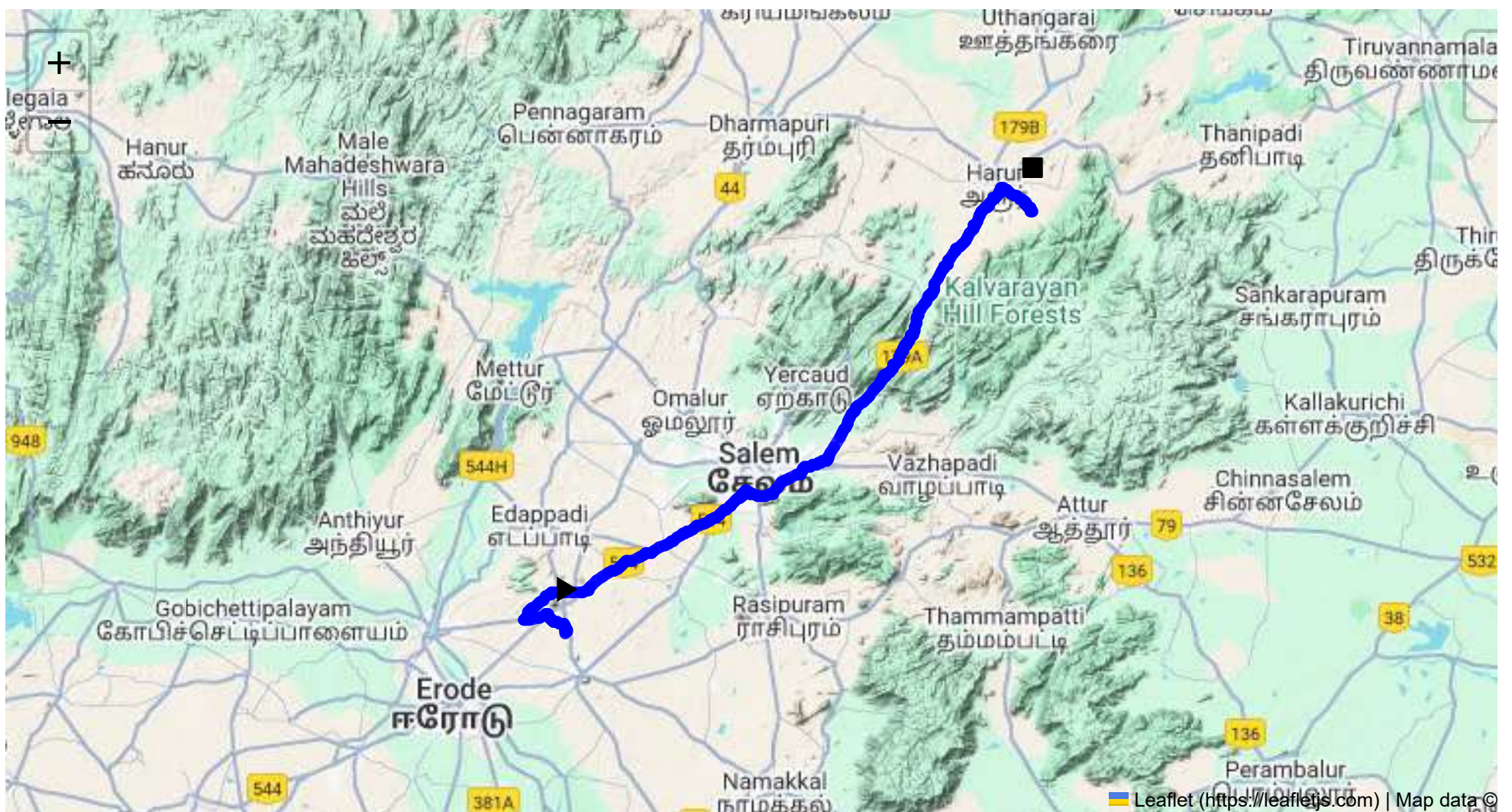
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 120.15 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.014748, 78.525917)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 120.15 kilometers, initiating at Sangagiri, Tamil Nadu, and concluding at Harur - Sitheri Road in Kelapparai, Tamil Nadu. Key waypoints include areas near Sankari, Salem, and Bhavani Main Road, utilizing major highways such as NH544. This route traverses both rural and semi-urban areas, each with unique infrastructure and traffic densities.

2. Typical Weather Conditions

The climate in Tamil Nadu is tropical, typically featuring hot and humid conditions. The region experiences heavy monsoons between June and September. Possible weather-related hazards include:

- **Monsoons:** Slippery roads and reduced visibility.
- **Summer Heat:** Potential for heat-induced vehicle issues.

3. Traffic Patterns

- **Peak Hours:** Morning (7-10 AM) and evening (5-8 PM) are peak traffic periods, especially around Salem.
- **Congestion Prone Areas:** Salem and its surrounding regions, particularly near urban intersections.

4. Assessment of Road Quality and Infrastructure

- **NH544 and Bhavani Main Road:** Well-paved, accommodating heavy vehicles but may have congestion.
- **Secondary Roads:** Potential for narrower lanes, variable maintenance levels, and possible potholes, requiring cautious navigation.

5. Suggestions for Alternative Routes

- **Emergency Detours:** Utilizing parallel state highways or smaller roads to bypass major congestion hubs, such as sidestepping around central Salem.

6. Local Regulations for Hazardous Material Transport

Tamil Nadu likely enforces national laws regulating the transport of hazardous materials, mandating:

- **Night Travel:** Regulations favor off-peak transportation to ease congestion.
- **Permits:** Essential for certain hazardous materials, ensuring compliance with safety standards.

7. Historical Incidents

- **Heavy Vehicle Accidents:** Incidents have occurred, particularly in high-congestion zones like Salem, often attributed to dense traffic and narrow urban roads.
- **Hazardous Material Spills:** Rare but possible in areas with tight roadways and challenging maneuvers.

8. Environmental Considerations

- **Ecosensitive Areas:** Avoid spillage near water bodies and agricultural zones to prevent contamination.
- **Wildlife Crossings:** Possible along rural stretches, demanding cautious driving.

9. Communication Coverage

- **Urban Areas:** Strong mobile network coverage near Salem and other connected towns.
- **Rural Dead Zones:** Potential patchy coverage, particularly in less developed stretches, making satellite phones a prudent backup.

10. Estimated Emergency Response Times

- **Urban Segments (e.g., Salem):** Typically fast, about 20-30 minutes.
- **Rural Roads:** Longer, potentially over an hour, due to distance and resource availability.

12. Overall Summary of Risk Assessment

This route encompasses a mix of highway and rural road conditions, with noteworthy risks including adverse weather, traffic congestion, and suboptimal road infrastructure in certain areas. Emergency response times vary widely, emphasizing the necessity for comprehensive pre-trip planning and abidance by regulatory protocols for hazardous materials. Despite its challenges, the route remains manageable with adequate preparation and situational awareness, prioritizing safety and responsiveness to evolving conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45571, 77.81702	30 KM/Hr
9	Turn	High	11.45615, 77.81677	15 KM/Hr
10	Turn	Medium	11.53552, 77.95763	30 KM/Hr
11	Turn	High	11.61233, 78.10397	15 KM/Hr
12	Turn	Medium	11.61235, 78.10495	30 KM/Hr
13	Turn	Medium	11.61422, 78.10810	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.65825, 78.20469	30 KM/Hr
15	Turn	High	11.66366, 78.20649	15 KM/Hr
16	Turn	Medium	11.67116, 78.23102	30 KM/Hr
17	Turn	High	11.67145, 78.23141	15 KM/Hr
18	Turn	High	11.67182, 78.23141	15 KM/Hr
19	Turn	Medium	11.67422, 78.23881	30 KM/Hr
20	Turn	Medium	11.68017, 78.24250	30 KM/Hr
21	Turn	Medium	11.83514, 78.35305	30 KM/Hr
22	Turn	Medium	11.99554, 78.44553	30 KM/Hr
23	Turn	Medium	12.04286, 78.48352	30 KM/Hr
24	Blind Spot	Blind Spot	12.04669, 78.48390	10 KM/Hr
25	Turn	Medium	12.01810, 78.52365	30 KM/Hr
26	Turn	High	12.01502, 78.52619	15 KM/Hr
27	Blind Spot	Blind Spot	12.01449, 78.52612	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
13	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
18	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
19	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
20	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
21	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
22	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
23	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
24	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
25	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
28	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
29	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
30	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
37	hospital	Government Hospital, Keeraipatti	12.0128824, 78.5263664	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
26	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
27	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
31	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
32	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
33	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
34	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
35	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
36	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53552, 77.95763



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61233, 78.10397



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61235, 78.10495



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61422, 78.10810



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65825, 78.20469



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66366, 78.20649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67116, 78.23102



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67145, 78.23141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67182, 78.23141



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68017, 78.24250



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83514, 78.35305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99554, 78.44553



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04286, 78.48352



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.04669, 78.48390



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.01810, 78.52365



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.01502, 78.52619



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.01449, 78.52612
